



Machine-learning-aided DFT-1/2 calculations for bandgaps of zinc oxide thin films

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ABSTRACT

Using the Density Functional Theory (DFT) -1/2 method, the electronic structure and bandgap of zinc oxide (ZnO) are calculated, providing the initial data sets for machine learning based on the genetic-algorithm-based artificial neural networks (GANNs) model. The predicted bandgaps from the well-trained GANNs model are close to (less than 5%) the calculated ones from the DFT-1/2 method along cutoff radius = 2.4 for Oxygen and cutoff radius = 1.2 for Zinc, agreeing with the experimental data. Our results show that combining the DFT-1/2 method with GANNs is an efficient way to correct the band gap of Zinc Oxide in DFT simulation.

1. Introduction

First-principles calculations based on density functional theory (DFT) have achieved a great progress in predicting and understanding the physical and chemical properties of materials [1–5]. However, prediction the bandgaps of several metal oxides, such as zinc oxide (ZnO), a promising material for solar-cell, photocatalytic and photo-degradation applications, is still challenging [6,7]. Although the DFT-1/2 method provides a possible approach to the end, as it functions in modeling the bandgaps of halide perovskites [8], CdTe, [9] and alloys [10] with high accuracy and relatively low cost, there remains an obstacle to overcome. Whereas the cutoff radius (R_{cut}) and the power index (PI) are required parameters for maximum the bandgaps, they strongly depend on the target material and do not follow a general rule. The process of finding suitable parameters thus could be costly. However, as genetic algorithm neural networks (GANNs) show potentials in developing emerging materials [11–15] including but not limited to identifying low-cycle fatigue behavior of materials [16] and designing components made of a multiphase material [17], it can help to find the parameters efficiently.

In this work, the electronic structure of wurtzite-type zinc oxide is calculated by the DFT-1/2 method, providing initial data for machine learning. The cutoff radius is selected randomly within the range of 0–5 bohr and the power index is 8, 20 or 50. Various GANNs models with different combination of hidden layers and neuron number are established and compared. The well-trained GANNs model generates 2600 optimized data points and shows the accuracy of prediction. The bandgaps of ZnO with optimized parameters by well-trained model are

calculated and in good agreement with experimental results.

2. Computational method

First-Principles Calculations were performed by using Vienna ab initio simulation package (VASP) with projector augmented-wave (PAW) method and local density approximation. The cutoff energy was set to 520 eV, and the convergence tolerances for the residual force and energy on each atom during the structural relaxation were set to 0.05 eV and 10^{-5} eV respectively. The Brillion zone integral was generated by the gamma method with a $(7 \times 7 \times 7)$ grid for k-points sampling. The DFT-1/2 method was used for bandgap correction. With the implemented of the method, the self-energy was multiplied by the truncated term presented in Eq. (1),

$$\Theta(r) = A \left[1 - \left(\frac{r}{R_{cut}} \right)^n \right] \text{ for } r \leq R_{cut} \quad (1)$$

$$= 0 \text{ for } r > R_{cut}$$

where the cutoff radius was explored to maximize the bandgap of target materials, the power index n was 8, 20 or 50, and A was the amplitude of the correction. The bandgap acquired from the calculation was the sum of the standard DFT bandgap plus the self-energy contribution [18–20].

GANNs are the combination of genetic algorithm (GA) [21–23] and artificial neural networks (ANNs) [24–26], comprising of an input layer, an output layer, and one or more hidden layers in the middle of them. In the hidden layers, the neurons in the adjacent layers are connected by

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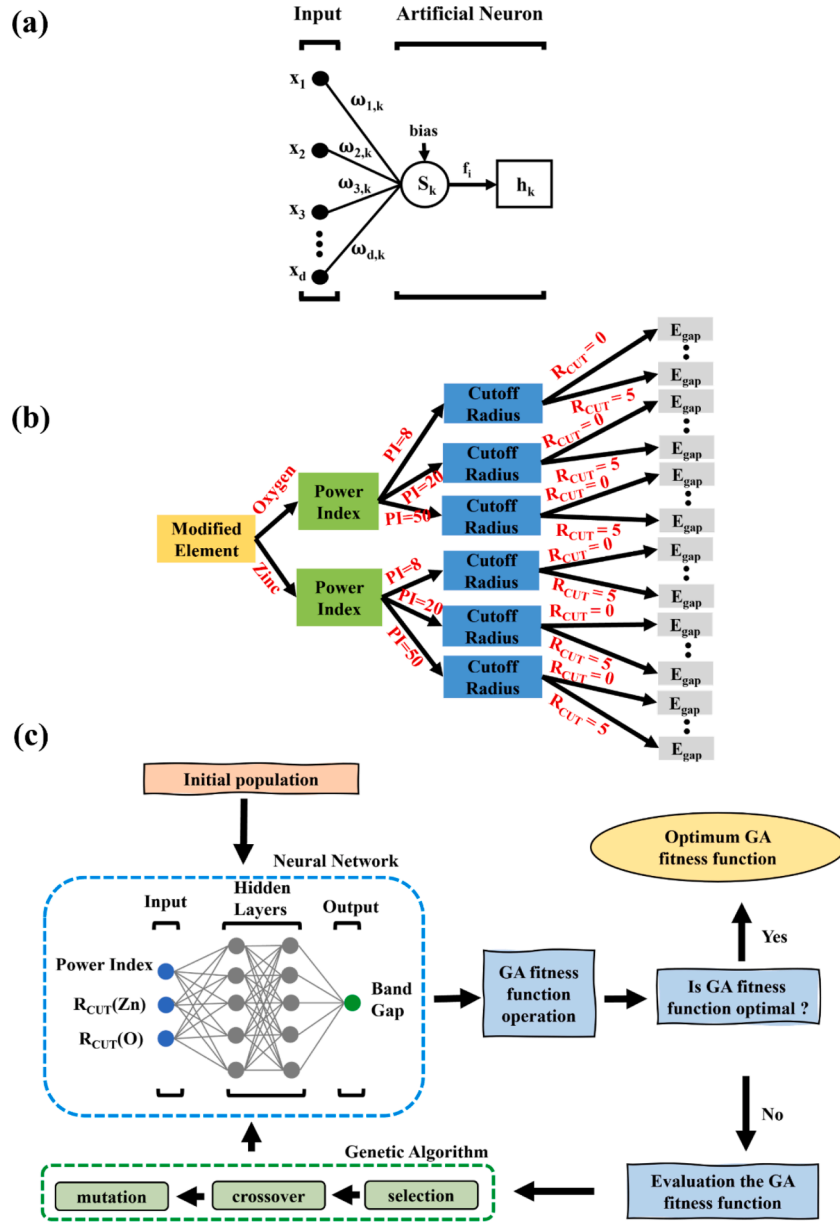


Fig. 1. (a) The neuron model. (b) Decision tree for parameter selection of the DFT-1/2 method. (c) flowchart of GANNs for the parameters of the DFT-1/2 calculations.

weights. A neuron model is showed in the Fig. 1a. In this model, the x_i is the i th input, the $\omega_{i,k}$ is the weight for the i th neuron, and f_i is the activation function.

S_k is defined as

$$S_k = \sum_{i=1}^d \omega_{i,k} x_i \quad (2)$$

And the output result h_k is calculated by

$$h_k = f_i(S_k + \text{bias}) \quad (3)$$

The artificial neural networks (ANNs) increase its accuracy in the training phase by back-propagation algorithm. The method feed the output error backward through the network and altering connecting weights to minimize the error between the network output and the target output. Through the altering of connecting weights for different inputs, the neural networks will learn to recognize the key parameter that determine the final output [27,28]. Once the data are input into the ANNs, the algorithm will learn and improve its accuracy by this method

to fit the nonlinear relation of the input data. Since the performance of ANNs highly depends on the quality of input data, it is an efficient method to improve the data with GA before machine learning starts. ANNs are expected to show better accuracy while training with optimized data [29–31].

For GANNs model assembly, the decision tree and flowchart of GANNs for bandgap prediction is shown in Fig. 1b, c. Step one is to input initial data generated from DFT-1/2 calculations into neural networks to train the machine learning model. After the operation, the model is judged by a fitness function to exam whether it is optimal or not. If the answer is yes, then the loop will stop. Otherwise, the process will go on to the section of GA. In genetic algorithms, each solution is represented as a string of binary numbers, known as a chromosome. These chromosomes are composed of genes while genes are composed of bits, which is the most basic unit of information in the computing. A bit is either 0 or 1. In the GA stage, the initial data endures selection, crossover, and mutation. In the initial part, data is transformed into chromosomes and then selected by the fitness function. The score of fitness

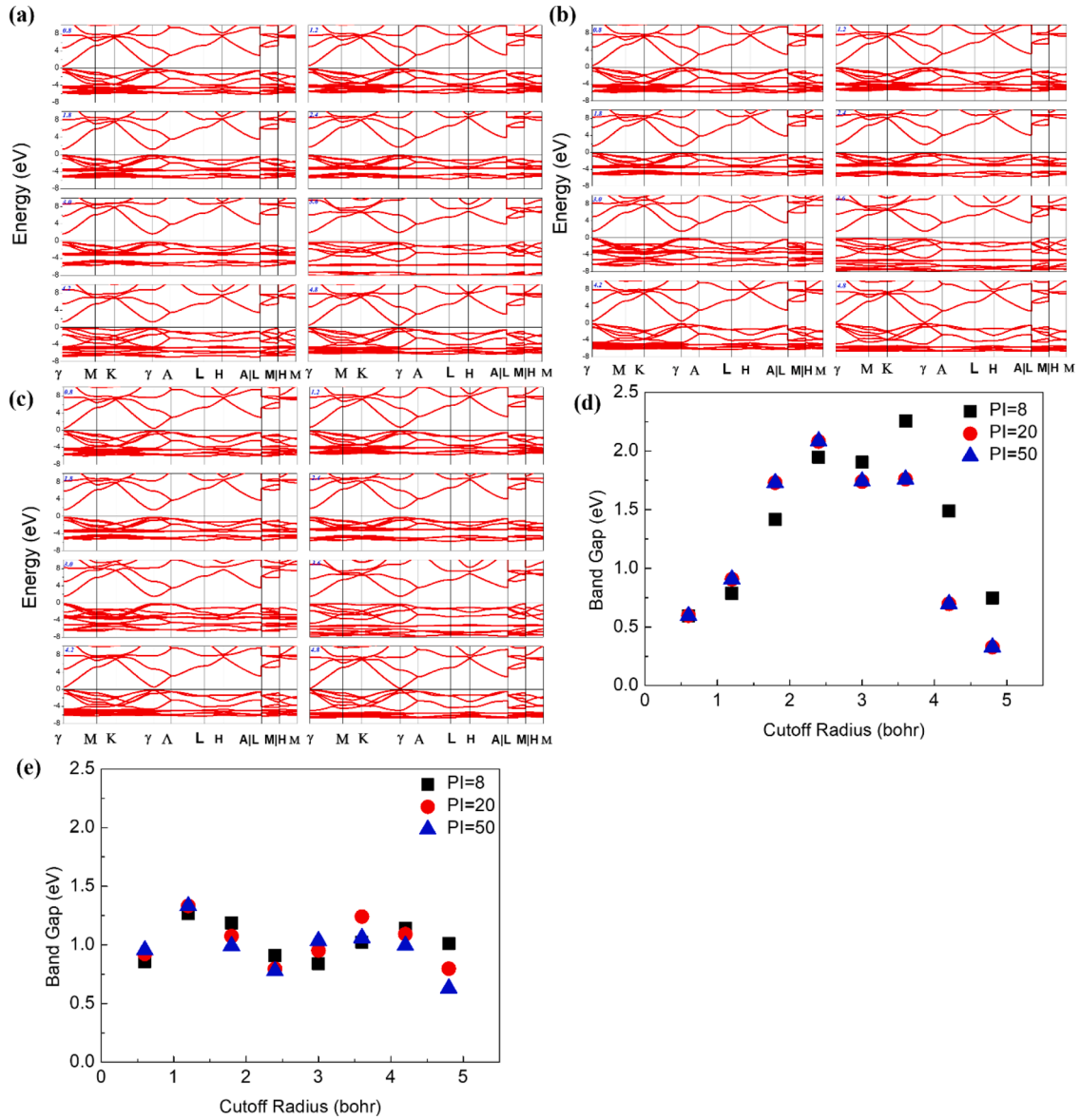


Fig. 2. The band structures of ZnO with various cutoff radius for power index = (a)8, (b)20, and (c)50. The bandgaps of ZnO with various cutoff radius and power indexes for (d) O and (e) Zn atoms.

function is defined as

$$fitness = \frac{1}{1 + RMSE} \quad (4)$$

RMSE is the root mean square error. It is defined as

$$RMSE = \frac{1}{n} \sum_{i=1}^n \sqrt{(Y_i - \bar{Y}_i)^2} \quad (5)$$

The Y_i represents the target output and the $\bar{Y}_i$ represents the solution output.

The smaller the difference between solution output and target output, the higher fitness score it will be given.

In the selection section, two chromosomes are chosen randomly. The chromosome with higher fitness score will be chosen as the parent in the next generation while the other one will be disposed. In the crossover phase, chromosomes exchange part of its genes with each other. Two chromosomes will exchange 70% of its binary number with each other to generate offspring. The crossover operation will increase the diversity of chromosomes and thus improve the possibility to find a better solution

for the problem in consideration. In the mutation section, 0.7% of bits in the chromosome will be change into the opposite one. The mutation inside chromosomes will disturb the solutions in the population pool, resulting in the opportunity to escape from the regional solution [32–34]. After the GA process, the generated data is input into ANNs again to produce well-trained model. The cycles continue until the model passes the GA fitness function test.

GANNs models with different hidden layers and neurons were trained with the DFT calculations. To acquire the desired well-trained GANNs model with best prediction accuracy, the RMSE of each model was compared. RMSE is commonly used in determining the performance of GANNs. A desire RMSE value should be near zero since that implied high-accurate prediction.

3. Results and discussion

Fig. 2a–c shows the calculated band structures of ZnO from the DFT-1/2 method for various R_{cut} and PI value. The pseudopotentials of the oxygen atom are modified by DFT-1/2 method. The bandgaps depend on the cutoff radius and more states are observed in the valance band for

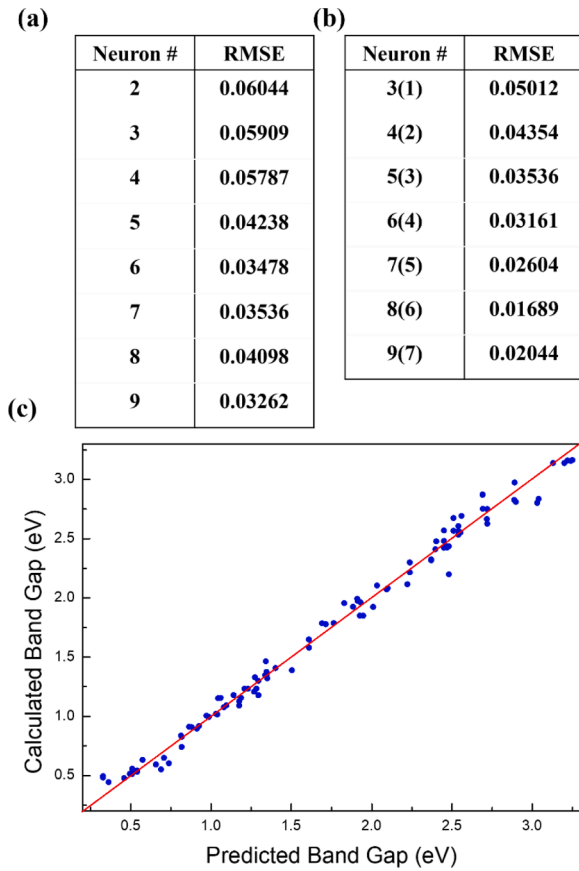


Fig. 3. The sorted RMSE of ANNs for (a) single hidden layer and (b) double hidden layer. (c) The result scatter diagram of GANN model with lowest RMSE. The $Y=X$ line represents ideal prediction.

cutoff radius larger than 3.0 bohr due to the non-degenerate states in larger cutoff radius. Fig. 2d, e show the bandgaps of ZnO, adjusted by the DFT-1/2 method on O and Zn atoms, respectively. Fig. 2d show that the maximum bandgap appears at the range cutoff radius from 2.5 to 3.5 bohr, consisting with the cutoff radius of 2.76 bohr for oxygen in nitrogen-doped TiO_2 [35]. Instead of a single peak, Fig. 2e reveals bimodal curves, suggesting that the maximum bandgap occurs at the cutoff radius of 1.2 or 3.6 bohr for the Zn atom. Nevertheless, none of the bandgaps are larger than 1.5 eV. Though the bandgaps of ZnO are opened by the DFT-1/2 method, they are still far below the experimental data (~ 3.2 eV) [36]. However, the parameters of the DFT-1/2 method can be optimized by machine learning for better bandgap calculations. Whereas the parameters are selected randomly, 148 calculated bandgaps are input into GANNs to train the machine learning models.

Fig. 3a, b show the RMSE values of GANN models for single hidden and double hidden layer. Neuron # stands for the number of neurons used in machine learning. For the double hidden layer model, the 1st number is the neurons in the 1st hidden layer while the number in the brackets represents the neurons in the 2nd hidden layer. Among all models, GANNs model with 8 (6) neurons in the 1st (2nd) layer demonstrates the lowest RMSE value (0.01689), selected as the well-trained model to be applied in bandgap prediction. The result-scatter diagram of GANNs well-trained model is shown in Fig. 3c. The ideal prediction results in $X=Y$ for all data points. The figure suggested that the well-trained model is indeed a reliable one since all data points are close to the $X=Y$ line.

2600 DFT-1/2 parameters for each power index are input into the well-trained model to find corresponding bandgaps. These parameters are the combination of different $R_{\text{cut}}(\text{Zn})$, $R_{\text{cut}}(\text{O})$, and PI. Both $R_{\text{cut}}(\text{Zn})$ and $R_{\text{cut}}(\text{O})$ are range from 0 to 5 while PI can be either 8, 20, or 50.

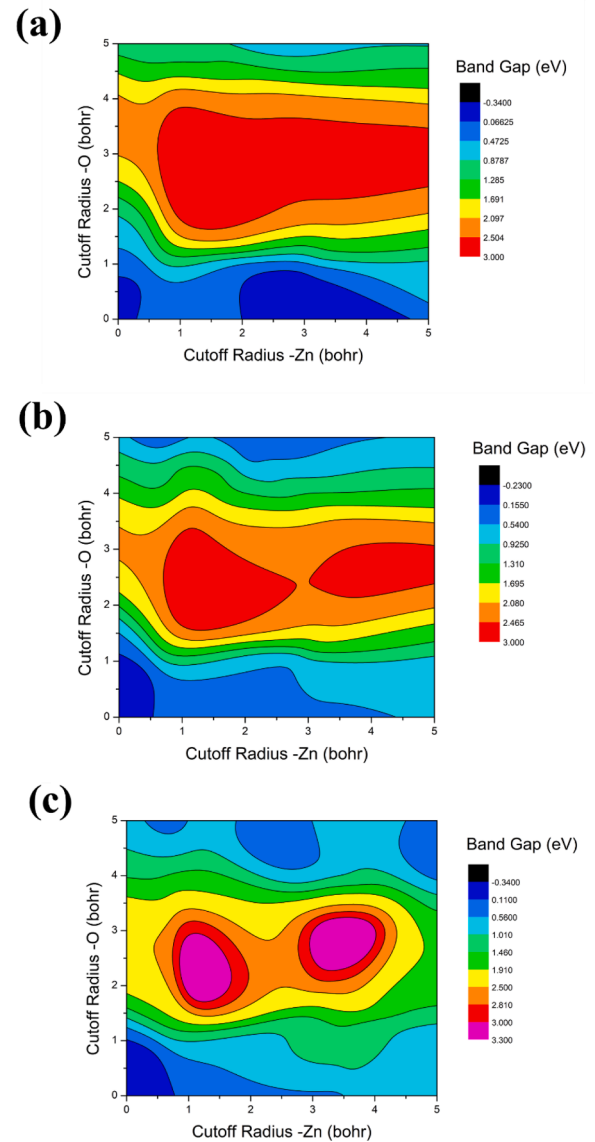


Fig. 4. Contour plots showing the distribution of 2600 data points generated from well-trained GANN model for power indexes of (a) 8, (b) 20 and (c) 50.

Once the parameters are input into the well-trained GANNs model, the algorithm will output the corresponding band gap for each individual parameter. As the result, the band gap values under almost all possible DFT-1/2 setting are acquired. Fig. 4a shows the contour plot of band gaps under $\text{PI} = 8$. The bandgap peak appears at the cutoff radius of oxygen around 2.5–3.5. On the other hand, the cutoff radius of zinc does not affect its bandgap in this range. The result suggests that oxygen dominates the bandgap of ZnO for power index of 8. However, no calculated bandgap close to the experimental one. Fig. 4b shows the bandgaps from power index of 20. Though the peak position still depends on the cutoff radius of oxygen, the influence of zinc emerges. The line along $R_{\text{cut}}(\text{O})$ is no longer flat. Two individual peaks appear at $R_{\text{cut}}(\text{Zn})$ of 1.2 and 4.8, and the former is larger. However, none of them reach 3.0 eV. Fig. 4c shows the bandgaps for power index of 50. Zinc contributes as the two peaks became higher, and the average bandgap increases. Several bandgaps are larger than 3.2 eV, matching with the experimental data. Overall, the contribution of zinc on the bandgap increases with the increase of power index. The maximum bandgap appears at power index of 50.

To verify the prediction of the GANNs model, the bandgaps of ZnO are calculated by DFT-1/2 with $R_{\text{cut}}(\text{O})$ of 2.4 and $R_{\text{cut}}(\text{Zn})$ of 1.2. These

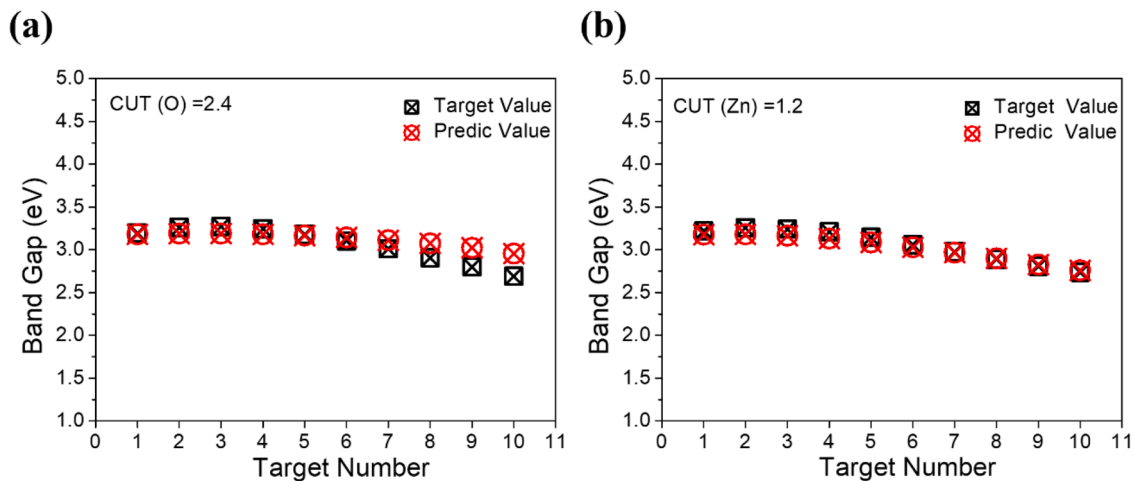


Fig. 5. The DFT $-1/2$ calculation outputs and GANNs outputs along the line. (a) $R_{\text{cut}}(\text{O}) = 2.4$ and (b) $R_{\text{cut}}(\text{Zn}) = 1.2$. Error is less than 5%.

two lines are selected because their predict bandgaps are similar to experimental ones. Ten points are selected in each line. Fig. 5 shows the comparison of calculations and GANNs outputs. The error between them is less than 5%, indicating that the GANNs machine learning can indeed aid to find the appropriate DFT-1/2 parameters effectively. Multiple possible combination of DFT-1/2 parameters can reach the experimental bandgap value. There are four points in Fig. 5a with bandgaps of 3.22, 3.25, 3.24, and 3.20 eV. In addition, three points in Fig. 5b show the bandgaps of 3.26, 3.27, and 3.24 eV.

4. Conclusion

The bandgaps of the ZnO are calculated by the DFT-1/2 method, which (148 data sets) are imported into GANNs for training the machine learning model. Double hidden layer with 8 neurons in the 1st layer and 6 neurons in the 2nd layer is the best choice. The well-trained GANNs model is capable of predicting the calculated results from the DFT-1/2 method with various parameters. Based on the generated data, the bandgaps of ZnO are larger than 2.8 eV when the cutoff radius of O is in the range 2~4 for all three kinds of power index. To verify the prediction of well-trained GANNs model, the bandgaps of ZnO with the DFT-1/2 adjustment along $R_{\text{cut}}(\text{O}) = 2.4$ and $R_{\text{cut}}(\text{Zn}) = 1.2$ are calculated, and the error between predicted and calculated bandgaps is less than 5%, in excellent agreement with experimental data. Our results thus show that combining the DFT-1/2 method with GANNs is an efficient way to calculate bandgaps.

CRedit authorship contribution statement

Wei-Che Tseng: Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Chao-Cheng Kaun:** Writing – review & editing. **Yen-Hsun Su:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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